Lab 7 – Trajectories

1 Introduction

Smooth motion increases precision of motion and reduces wear and tear on a robotic manipulator. In this lab a smooth motion trajectory system will be developed for the control of robotic simulator movements.

2 Experiment

Implement a smooth joint trajectory method and Cartesian trajectory method for the robotic simulator.

A) Modeled on the Robotics toolbox, add a ***jtraj*** polynomial function to the robot simulator. The inputs include an initial and final position, initial and final velocity (the initial and final acceleration = zero). Also include a variable for the number of time steps. Store the output of the function in a vector.

B) Add an option to use the joint trajectories algorithm to cycle between joint values (0, 0, 0, 0) and (-180, 90, 90, 15 cm) and then back to (0, 0, 0, 0).

C) Add a Cartesian trajectory method to the robot simulator. Implement the following algorithms:
   a. The initial trajectory location is the current end effector current pose, and the target position is the position and orientation from marker ID = 50. Use the ***jtraj*** method to interpolate values for X, Y, θ between the two positions and for each step, use ***ikine*** to determine the joint angles and display.
   b. Place 3 markers on the board and smoothly interpolate between all three, continuously track the markers so should one be moved the system will still track the correction position. A state machine may be useful. Use a fixed duration between moves or use the distance between current and target to scale the speed of the movement (further away = faster move).

